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Spatio-Temporal Analyses for Dynamic Urban Road Network Management

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Abstract

The dynamic urban network management in urban areas is in high demand, since speed information is currently available via Information and Communication Technology (ICT) Infrastructure. However, such information has not been incorporated into routing decision making systems efficiently. Although several efforts exit, these are not correlated and integrated. Since the integration problem has spatial and temporal dimensions, Spatial Information Science could aid to solve this problem via mature methods, approaches and tools. The main aim of this study is to demonstrate the opportunities of spatio-temporal analyses within dynamic urban road network management, where the concepts are supported with a case study from Istanbul Metropolitan Area. In the proposed methodology, the speed information retrieved from detectors were associated with the OpenStreetMap and quickest paths were simulated within determined time intervals. Different shortest routes are proposed to users in various time spans via incorporated speed information and network analyses. For the same origin and destination, 2.5 minutes difference is observed within the study area. The results were promising, where the added value of this study is presenting an approach for integrating various spatial-temporal data to aid dynamic urban road network management.

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Keywords: traffic speed information; Goegraphic Information Systems (GIS); Traffic; dynamic routing.

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1. Introduction

The need for dynamic route management is apparent, however current traffic management systems are not mature enough to accomplish it. The reason for such a problem is mainly depends on; complexity of the urban network, insufficient real-time data collection system and non-integrated decision support systems for traffic routing problem. Some efforts exit, where nowadays several service providers for selecting the best-routing option for destinations are available. However, these provide partial solutions, since the traffic management efforts are uncorrelated and not integrated. Additionally, this issue could only be solved via the cooperation of citizens and informed decision making. Since the traffic related information is available nowadays via Information and Communication Technology (ICT) Infrastructure, the solution for such a spatial problem should be dynamic as well. Hence, it is a spatio-temporal problem, where Spatial Information Science could aid to solve this problem via mature methods, approaches and tools.

The main aim of this study is to demonstrate the opportunities of spatio-temporal analyses within dynamic urban road network management, where the concepts are supported with a case study from Istanbul Metropolitan Area. The designed methodology is processing and analyzing speed information on hourly bases for week days and week-ends, where different shortest routes are proposed to users via network analyses. The well-known algorithm of Dijkstra is used to solve the shortest path problem for a connected digraph with non-negative weights. For a given source node in the graph, the algorithm finds the shortest path between that node and every other. For the network distance calculations, topological elements of the network data are used to perform the distance calculations from the origins to destinations, where for this study speed- time spend- is an additional parameter for the calculation.

This paper is organized as follows. The following section is background that illustrates some relative significant studies and works. The third section presents shortly important facts regarding the study area, used data and followed methodology. Finally, in the fourth section the obtained results are presented and general comments and conclusion are given.

2. Background

The major challenge presented in dynamic traffic management is the optimal utilization of all the routes according to their limitation in number and capacity. In most cases, as Shashikiran (2011) stated, it is inconvenient to construct new routes or increase roadway capacities. So identifying ways to maximize the utilization of the existing transportation network becomes more important. An effective way to do that is through Dynamic Routing (or adaptive routing) which has been intensely studied and investigated by road network designers and managers. Designers and managers use the Dynamic Routing to examine the capability of road networks to find alternative routes in response to a change in the system. Mak (2011) defines Dynamic Routing as a utilization of the online communication patterns and real-time information to effectively avoid overcrowding or faulty components and reduce the possibility of packets being continuously blocked, thus it helps minimize congestion and avoids hot-spot areas for road network users. Dynamic Routing within urban areas requires dynamic simulations, where the emerging information and communication technologies (ICT) could generate new sources of data and opportunities. However, such data requires new approaches for data acquisition, modeling, analyses and visualization. Recently, real-time traffic data is available for almost all cities, however, such information is not incorporated into our daily transportation routines.

Several studies have been performed trying to dynamically find the best routing system in road networks to, as possibly, avoid traffic congestions and hot-spots. In this context, traffic flow models and algorithms were developed to predict the traffic congestion occurrence in the networks. For that, free traffic congestion road networks were needed to be designed and managed. In order to do such things, various significant data of road network components were collected and analyzed by different methods and tools gradually improving over time. Here we display, shortly, some relative studies in this direction. Leblanc and Abdulaa (1979) presented a computationally efficient technique for determining the optimal design of an urban road network, by assigning of network flows and the determining of improved link parameter values so that congestion is minimized. Both of Wang et al. (2005) and Wang et al. (2009) developed general approach to the real-time adaptive of the complete traffic state in freeway routes or networks, based on the stochastic nonlinear macroscopic traffic flow modeling and extended Kalman filter. Cantarella et al. (2006)

evaluated the performances obtained from various well known algorithms proposed for the real networks topology design stage. Chang et al. (2012) described some interesting traffic prediction and collection technologies deployed in Taiwan, such as vehicle detector, GPS-based vehicle probe and cellular floating vehicle probe, and specified their intended purposes. Chang et al. (2014) described fusion technique to integrate active and passive data from spot and space data for the estimation of traffic speed in emergency scenarios based on entropy and optimal weight. Dynamic Origin-Destination (OD) is one of the most important estimation methods in achieving the potential promised by the self-adaptive traffic control system for urban traffic management. This method was implemented by Jingxin et al. (2014) using multi-source sensors data. Wang et al. (2015) developed an urban road network traffic state dynamic estimation method to be used in the design of urban network traffic state and phase estimator. Charansonney and Aguiléra (2015) proposed a traffic dynamic macroscopic model to study how different classes of users interact on road networks, each class being characterized by the type and level of information it has access to. Falcocchio and Levinson (2015) illustrated how operational strategies can reduce congestion by making better use of existing streets and highways. Wongdeethai and Siripongwutikorn (2016) proposed a road traffic collecting protocol to gather road traffic information in a vehicular ad hoc network on city roads. Raiyn (2016) introduced a short-time forecasting model based on real–time travel time for urban heterogeneous road networks.

3. Data and Methodology

In this section, the case study area, data used and the methodology is explained briefly. The study area is located inside the Istanbul Metropolitan area, where especially urban transport is considered as one of the major problems. As a result of investment policies focusing heavily on roads, while the arable lands and forests are vanishing, the unplanned increase of population and industrial facilities along the network were increased.

According to municipality public transport authority -Istanbul Electricity, Tramway and Tunnel Establishments-(IETT), the public transport trips for 2014 is 2.5 million/day, where cars passing bridges are 53 million per year. The total road network is 30.000km for all provinces, where the municipality is responsible for 4.000 km (Fig. 1). Within the Istanbul metropolitan area, the developed concepts are implemented in the road network of Bahçelievler district, where the area is illustrated inside the black box in Fig. 1. The previous numbers suggest how Istanbul suffers badly from road traffic. The lack of coordination between transport policies, land use policies and environmental policies caused a growth of an expensive and polluting transport system even for Istanbul, which has a well-linked multi-modal transport system. Long commuting distances and trip hours were observed, especially at peak hours.

The spatial data for dynamic routing analyses are mainly OpenStreetMap road network data and location of speed detectors of Istanbul Metropolitan Municipality. As the base map for road network, the OpenStreetMap navigation road network is used that is freely available under (https://www.openstreetmap.org). Arterial roads, junctions and highways are selected from the Openstreet map road network database. The 6 speed detectors located in the study area are used. The spatial distribution and the other attributes of the traffic speed detectors are presented in Figure 2 and Table 1. The average speeds of the vehicles using each network element in the road network were registered to the database using speed detectors belonging to the Istanbul Metropolitan Municipality.



Fig. 1. Arterial roads of Istanbul's road network. Bahçelievler's road network inside the black box.

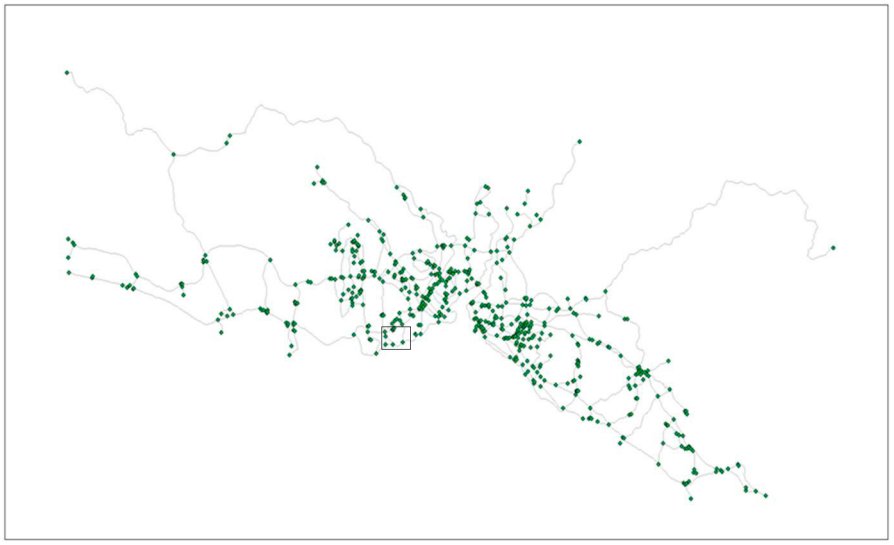


Fig. 2. Positions of speed detectors in Istanbul's arterial roads and in the test area inside the black box.

Table 1. Illustrates the name, location and areas each used detector connects.

Detector's name	Direction Count	Lane Count	From	To	X-coord.	Y-coord.
D100 Şirinevler	2	6	Sefaköy	Bakırköy	28.85323409	40.99210206
Yenibosna	2	6	Merter	Florya	28.8335434	40.99273287
Bakırköy	2	6	Çobançeşme	Merter	28.86911335	40.99682415
B. Ekspres Yolu	2	6	Havaalanı	İstoç	28.81246155	41.01324549
D100 Türk Böbrek Vakfı Önü	2	7	Merter	Bakırköy	28.87769559	40.9995373
Kuyumcukent Basın Ekspres yolu	2	5	Çobançeşme	İkitelli	28.81925762	40.99855696

Seven time spans of the day were determined (i.e. 00-07, 07-09, 09-12, 12-15, 15-18, 18-21 and 21-24), based on the traffic usage. The average speed is registered on a 15 minutes interval (Table 2), later it's averaged for each hour and finally for each interval as illustrated in Fig. 3. The average speed varies in each road around day time, less traffic congestion allows road users to drive faster.

Table 2. Sample illustrates the average speed for each direction in 15 min interval for the detector of Yenibosna.

Date -Time	Speed in direction 1 (Km/hour)	Speed in direction 2 (Km/hour)	From	To
1/24/2016 - 0:00	63	76	Merter	Florya
1/24/2016 - 0:15	67	78	Merter	Florya
1/24/2016 - 0:30	73	83	Merter	Florya
1/24/2016 - 0:45	78	84	Merter	Florya
Average speed	70.25	80.25		

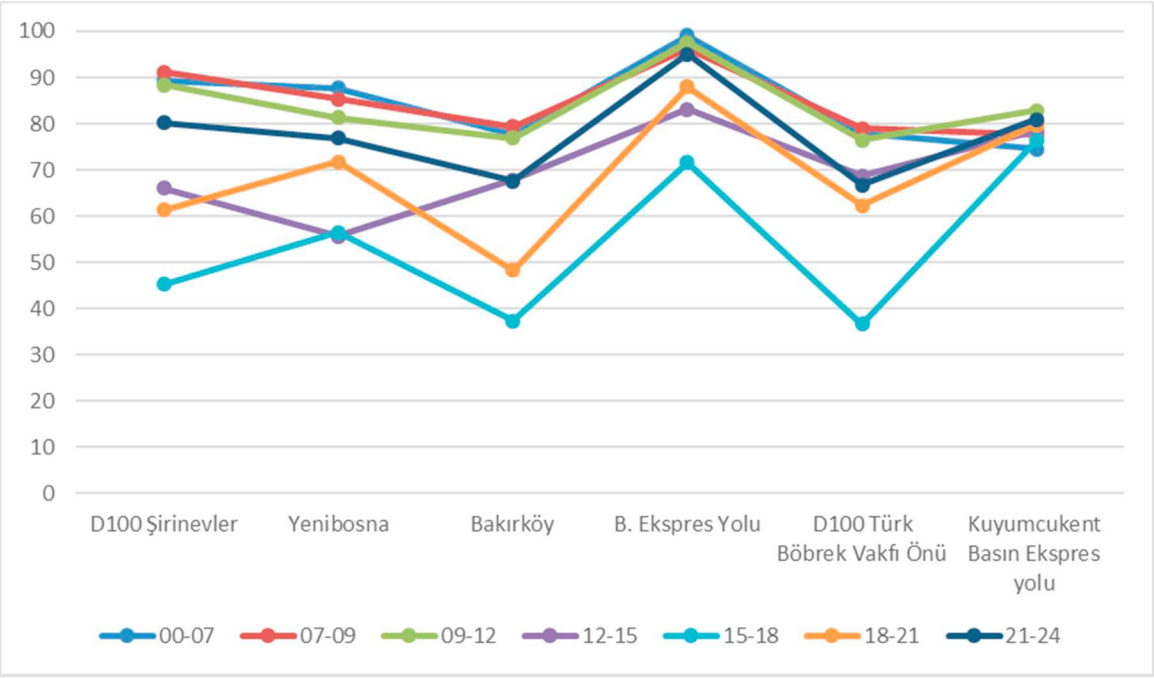


Fig. 3. Chart illustrates the speed (Km/hour) for each time interval in its corresponding detector

The problem of finding shortest paths from origin node (vertex v) to all other vertices (destinations) in the graph is a well-known, and forms the bases of graph theory. A simple directed graph G is comprised of {V;E}. The V is a set of vertices and E is a set of edges. Edges are binary relationships between vertices (u ; v) such that u; v in V and $u \neq v$. A path between a source vertex v_1 and a destination vertex v_n in V is a sequence of adjacent vertices $\{v_1; v_2; .....; v_n\}$. If a graph G is connected then $\forall v_i; v_j$ in V there exists a path between v_i and v_j (Merrifield 2010). The time – dependent graph (G') utilizes a cost function (e; t), where t is the current time. A time –dependent graph can also be referred to as a dynamic graph.

A well-known algorithm to solve the shortest path problem for a connected digraph with non-negative weights is Dijkstra (1959)'s algorithm. For a given source node in the graph, the algorithm finds the shortest path between that

node and every other. This algorithm examines the connectivity of a network to find the shortest path between two vertices. For the network distance calculations, topological elements of the network data are used to perform the distance calculations from the origins to destinations. Distances are measured through the route, not as the crow flies.

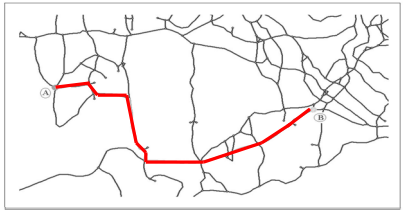
4. Results & Conclusion

A sample origin-destination points were selected to analyze the results. The shortest path without the speed information between A and B is presented in Fig.4.a. When the time parameter was introduced, two different routes between A and B were calculated for different time slots. The route presented in Fig.4.e is 2.5 minutes shorter than the previous route. The alternate route continues to be the quickest path in the time span 12.00 and 21.00. After 21.00, again the quickest route will be the previous route as presented in Fig.4.h. After calculating the possible routes based on real-time spatial data, alternatives were simulated and such information could be used to assist decision makers

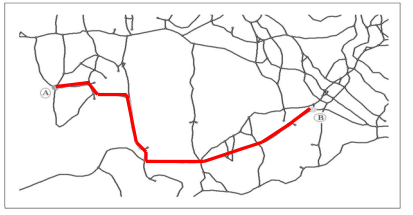
In order to estimate the significance of these 2.5 minutes, the approximate cost for this delay should be calculated. According to Christidis and Rivas (2012) the cost of road congestion in Europe is over €110 billion (122 billion $) annually.

For the case study area, Istanbul Metropolitan Area, it was calculated as for each person 3.57 $/hour. (ITRC 1997) Recent analyses reveals that, the delay time at traffic for Istanbul costs nearly 1.8 billion $/year. According to the Istanbul Transportation Master Plan-2011 (IBB 2011), the traffic volume of these routes is 225.000 person/hour. A coarse estimate of losing 2.5 minutes could be calculated as approximately 4.8 million $ per year that could be saved if the quickest path was used.

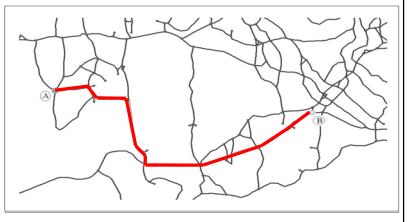
The selected the study area, the road network is relatively dense compared to the other districts of Istanbul as illustrated in Fig.5. Other connectivity parameters for the study area, such as betweeness of a node: 0.004, detour index:0.0008, geodetic distance 33145.53, are also in line with this “medium connected” classification. The quickest path analyses should also be conducted for sparse and dense road networks and multi-modal networks, where results could aid more collaborative decision making on dynamic routing.



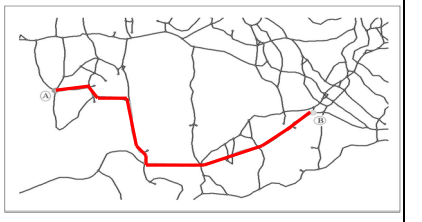
a) shortest path calculation without time(t)



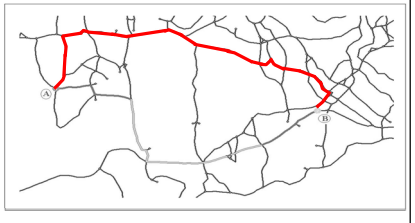
b) Route for 00.00-07.00



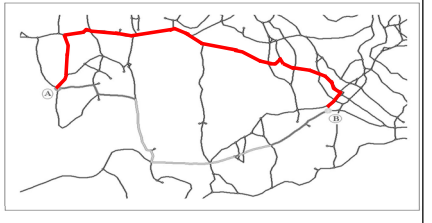
c) Route for 07.00-09.00



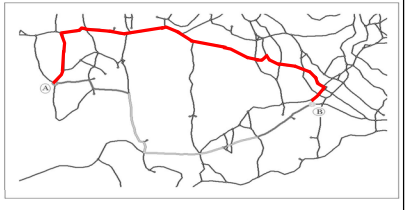
d) Route for 09.00-12.00



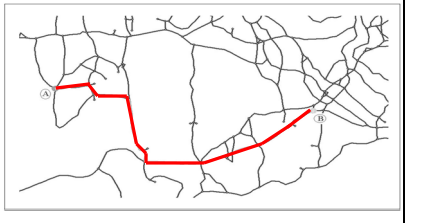
e) Route for 12.00-15.00



f) Route for 15.00-18.00



g) Route for 18.00-21.00



h) Route for 21.00-24.00

Fig.4. The quickest routes via dynamic urban road network

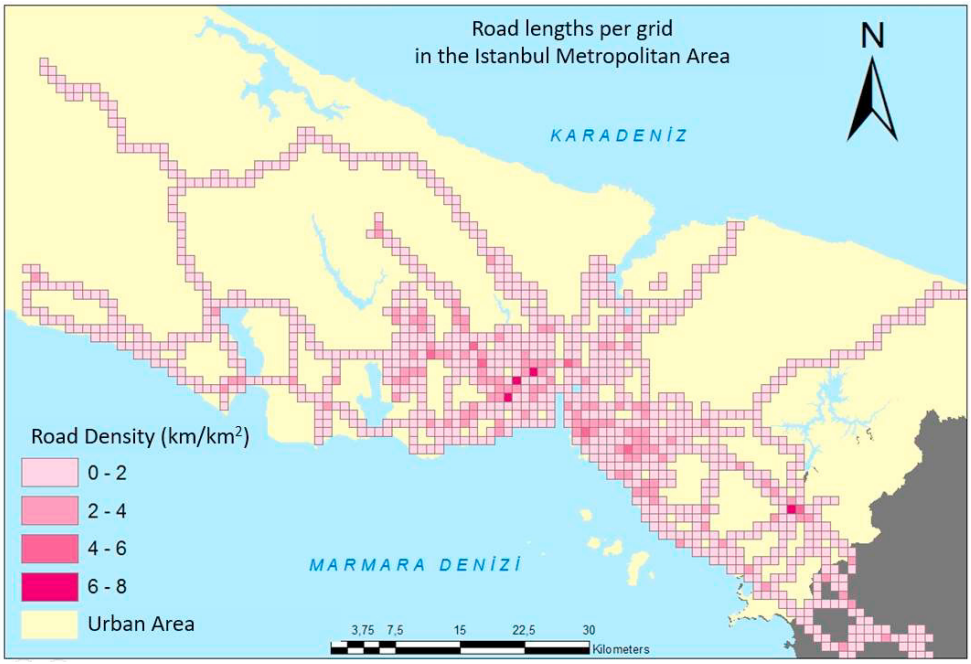


Fig.5. Relative road density of the Study area.

In this paper, the followed methodology allowed registration of single average speed around speed detector. Based on the used methodology, the shift between the neighbouring areas of each detector comes with a sudden change of average speed as illustrated in Fig.6, which is a coarse approach to the dynamic routing problem. The change in speed happens gradually as presented in Fig.7. The uncertainty involved within this estimation could be revealed via various techniques. In order to solve the problem of estimating the state of discretized hyperbolic scalar partial differential equations, macroscopic and microscopic traffic models are suggested and implemented.

A macroscopic traffic model considers traffic in the level of density and flow, so traffic is treated as streams. On the other hand, a microscopic model considers traffic in the level of individual vehicles, thus it models the dynamics of every single vehicle. Compared with microscopic models, macroscopic models have the advantage that it usually has fewer parameters to calibrate. Also, to calibrate or validate a macroscopic model, it only requires traffic data like flow, density or occupancy. This type of data can be collected easily by loop detectors. However, calibrating a microscopic model requires trajectories of individual vehicles, which has to be obtained through camera or GPS. Thus, macroscopic models have a larger source of available data and it is easier to access. Further, macroscopic models can be computed faster and take less computer resource. However, tests show that the inherent randomness of traffic flow and uncertainties in the initial conditions of models, model parameters, and model structures all influence traffic state estimations. On the other hand, microscopic models provide more accurate simulations of traffic flows through road networks.

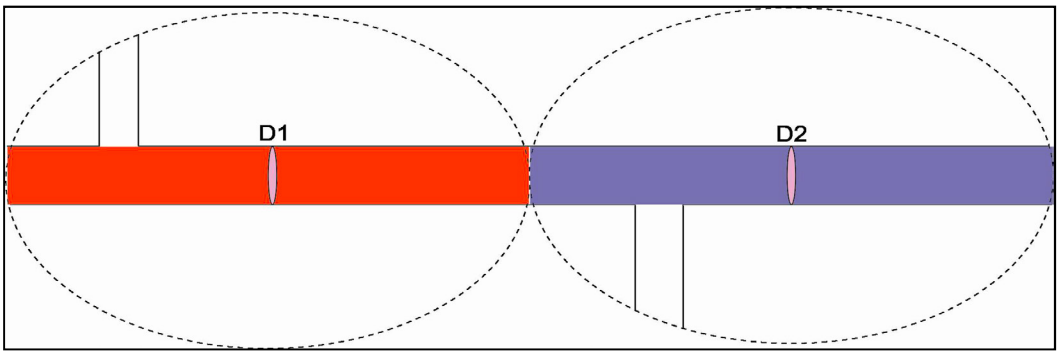


Fig. 6. Sudden change of speed using the proposed methodology. Red color represents average speed around detector D1, violet color represents average speed around detector D2.

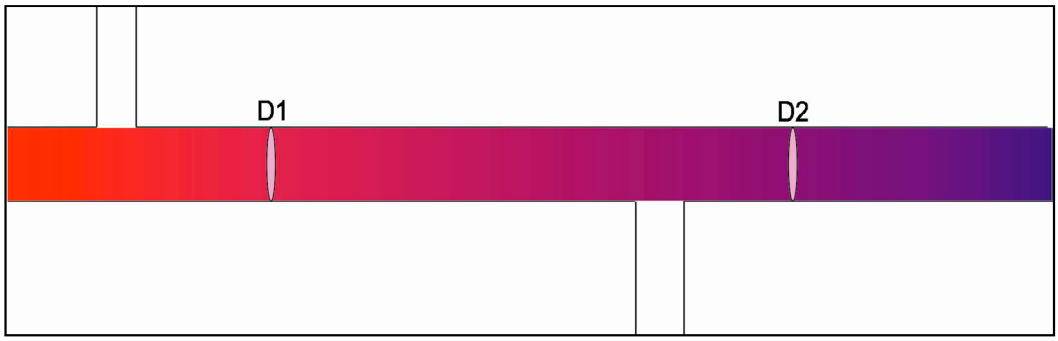


Fig. 7. Graduate change of vehicles' average speed represented in graduate change in color.

Several studies to estimate the speed information in street networks using macroscopic traffic models, such as Su et. al. (2013), Celikoglu (2014) and Li and Zhang (2014). Su et al. (2013) used the Cell Transmission Model (CTM) on a segment of urban street, where they proved the success of CTM model on a segment of signalized arterial. Celikoglu (2014) proposed a dynamic approach to specify flow pattern variations, simulated by a multimode macroscopic flow, and later incorporate the neural network theory to reconstruct real-time traffic dynamics. Li and Zhang (2014) presents an ensemble learning framework to appropriately combine estimation results from multiple macroscopic traffic flow models. On the other hand, Herrera and Bayen (2009), Khondaker and Kattan (2015) and Quddusa and Washington (2015) implemented interested microscopic traffic model. Herrera and Bayen (2009) used cell-phones equipped with a global positioning system (GPS) to accurately provide position and velocity of the vehicle as probe traffic sensors. Khondaker and Kattan (2015) presented a variable speed limit control algorithm for simultaneously maximizing the mobility, safety and environmental benefit in a connected vehicle environment. Quddusa and Washington (2015) developed a new weight-based shortest path and vehicle trajectory aided map-matching algorithm that enhanced the map-matching of low frequency positioning data on a road map to better estimate link travel time and speed from low frequency GPS data. The microscopic and macroscopic traffic models should be tested in order to diminish the uncertainties on speed data, where such analyses will be tested for the study area in further steps.

This work is still in progress and conclusions are, at this stage, tentative. However, significant differences on quickest paths in different time spans were observed. The added value of this study is presenting an approach for integrating various spatial-temporal data to aid dynamic urban road network management. Furthermore, speed data is incorporated into route optimization that could aid to reduce congestion at urban areas. The findings presented in this study suggest that spatial information, methodologies and approaches could generate useful results that could be used for collaborative decision making within the dynamic routing in urban road networks.

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